Incremental BERTopic and Distributional Drift Detection for Social Media Brand Crisis Monitoring

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Abstract

We study how to detect emerging social media crises for brands by combining transformer-based topic modeling with distribution-based change detection. Using a year of Twitter data for three brands (Waffle House, MARTA, and the Falcons), we train and tune a GPU-accelerated BERTopic model (RAPIDS cuML UMAP+HDBSCAN) with a custom Optuna objective that balances coherence, diversity, outlier management, and avoidance of "mega" clusters. To support near–real-time use, we compare two incremental strategies: (1) an online BERTopic variant based on IncrementalPCA, MiniBatchKMeans, and OnlineCountVectorizer, and (2) BERTopic's merge_models pipeline, which merges new per-window topic models into a stable baseline. The online variant serves as an auxiliary streaming baseline but suffers from unstable topic identities. Our primary method, merge_models with a global remapping layer, maintains topic stability while introducing interpretable new negative topics. On top of the merged model, we track topic and sentiment distributions via Jensen–Shannon divergence and negative-sentiment z-scores, yielding a practical framework for crisis monitoring at Sprout Social's scale.

1 Introduction

1.1. Motivation: Crisis Detection in Social Media

Social media gives every user a global publishing channel, transforming politics, culture, and commerce. For brands, this creates both opportunity and risk: it enables powerful marketing and customer care, but also allows negative narratives to spread rapidly.

Sprout Social serves more than 30,000 brands with analytics and management tools [1]. One key product line, *Listening*, helps brands understand trends, sentiment, and conversation topics across platforms, but stakeholders identified a key gap: crisis detection—alerting a brand when conversation content or sentiment shifts sharply toward negative themes such as service failures, safety issues, or scandals.

Manual monitoring is infeasible; even mid-sized brands receive tens of thousands of posts per week. The central challenge is to automatically structure this text into interpretable topics and detect meaningful shifts in the topical and sentiment landscape that may signal an emerging crisis [5].

1.2. Problem Setup and Data

Sprout Social's production environment ingests continuous, multi-platform, multi-lingual streams under latency and cost constraints. In this practicum we focus on a static but realistic subset: English tweets collected between July 2024 and August 2025 for three representative brands: Waffle House (n = 238,178), MARTA (n = 13,309), and the Falcons (n = 321,096). Each tweet contains de-identified text, a brand label, a timestamp, and a sentiment label (negative / neutral / positive) from Sprout Social's upstream classifier.

Our goal is not to design a full production system but to develop and evaluate the core modeling components for crisis detection:

1. a topic model that produces coherent, diverse topics with explicit outlier handling,
2. an incremental strategy that extends this model to new time windows without destroying topic identities, and
3. a change-detection module that converts daily topic–sentiment dynamics into interpretable alerts.

We keep the text-modeling stage self-contained: each tweet is assigned to either a topic or an outlier and retains its brand and sentiment labels. All downstream logic then operates on structured time series of topic and sentiment distributions, making it easy to plug in generic time-series and divergence-based tools.

1.3. Overview of BERTopic

BERTopic combines transformer-based sentence embeddings, dimensionality reduction, clustering, and class-based TF-IDF into a single pipeline with a simple *fit_transform* interface [6]. In our setting:

- we encode tweets with the *e5-base-v2* sentence embedding model, which is well suited to short, noisy social media text [12, 5];
- we reduce dimensionality with GPU-accelerated UMAP from RAPIDS cuML to preserve local neighborhoods efficiently [7, 9];
- we cluster with HDBSCAN (also via cuML), which allows non-spherical clusters and explicit outliers [3];
- we derive topic descriptors using c–TF–IDF and refine them with a representation model combining PartOfSpeech, KeyBERTInspired, and Maximal Marginal Relevance (MMR) [6, 4].

BERTopic additionally exposes *update_topics*, *merge_models*, and a restricted *partial_fit* API. These hooks let us explore incremental strategies built on top of a tuned baseline model.

2 Data, Metrics, and Tuning

2.1. Data Preparation

We apply standard preprocessing steps before topic modeling to convert the de-identified tweet text into clean, analysis-ready sentences. This includes removing retweets and quote-tweet boilerplate, normalizing whitespace and punctuation, lower-casing text, and retaining the associated brand and sentiment labels for downstream analysis. We divide the full dataset into a 70/15/15 time-based split and then further partition the test block as follows:

- **Training set** (70% of all data): used for Optuna-based hyperparameter tuning.
- **Validation set** (15% of all data): used for model evaluation during tuning.
- **Test set** (15% of all data): reserved for incremental modeling, further split into:
  - **Test-fit set** (70% of test set): used to extend the topic model to more recent data.
  - **Test-monitor set** (30% of test set, approximately 20,000 tweets): used as a proxy for a live monitoring window.

2.2. Embedding and GPU Acceleration

We pre-compute embeddings for all data splits using *e5-base-v2*, obtaining one dense vector per tweet [12]. These embeddings feed directly into UMAP and HDBSCAN, both instantiated from RAPIDS cuML [7, 9]. Running dimensionality reduction and clustering on GPU is essential: the search over BERTopic configurations involves repeated fits on hundreds of thousands of tweets.

Empirically, GPU UMAP+HDBSCAN reduces wall-clock tuning time from hours to tens of minutes on our search space. This makes it feasible to explore a richer set of hyperparameters and is consistent with Sprout Social's requirement to scale to many more brands.

2.3. Scoring Metrics and Objective Function

We define several metrics to capture the quality and usability of a topic model:

Coherence (coh): We compute topic coherence with a multi-topic variant of c_v (via *coherence_multi*). Higher coherence means top words in each topic co-occur in the underlying documents, which correlates with human interpretability.

Diversity (div): We measure topic diversity as the proportion of distinct words across all topics' top-N words. Higher diversity discourages near-duplicate topics that differ only by small phrasing variations.

Outlier rate (out): We track the proportion of documents labeled as outliers by HDBSCAN, both before and after calling BERTopic's *reduce_outliers*. We blend the two:

$$out_{\text{blend}} = \alpha \cdot out_{\text{after}} + (1-\alpha) \cdot out_{\text{before}},$$

with $\alpha = 0.7$. Very high outlier rates discard usable data; very low rates often mean HDBSCAN collapses weak clusters into large, heterogeneous groups.

Mega-cluster share ($mega$): We define $mega$ as the fraction of documents in the largest non-outlier topic. When $mega$ is large, the model effectively forms a single "catch-all" cluster, which is undesirable for crisis detection because emerging minority topics become masked.

We then define a scalar objective:

$$J = w_{\text{coh}} \cdot coh + w_{\text{div}} \cdot div - w_{\text{out}} \cdot out_{\text{blend}} + w_{\text{mega}} \cdot (1 - mega),$$

with weights

$$w_{\text{coh}} = 0.35, \quad w_{\text{div}} = 0.15, \quad w_{\text{out}} = 0.20, \quad w_{\text{mega}} = 0.30.$$

These weights reflect a balanced design choice informed by preliminary experiments and the needs of the topic-modeling pipeline. Within each Optuna trial, we fit the model on training set, compute metrics on both training and validation sets, and then use the validation objective J_{valid} as the scalar value returned to Optuna. Training metrics are recorded for diagnostics, but the selection of hyperparameters is entirely driven by J_{valid}.

2.4. Optuna Search Space and Design

We use Optuna [2] to tune both text and manifold/clustering hyperparameters of BERTopic. The key search dimensions are:

- CountVectorizer [8]:
  - n-grams: $(1,1)$ vs. $(1,2)$,
  - *min_df* $\in [0.003, 0.01]$ (log-scale),
  - *max_df* $\in [0.85, 0.95]$.
- UMAP:
  - *n_neighbors* $nn \in \{20, 30, \dots, 100\}$,
  - *n_components* $nc \in \{20, 40, 60, 70\}$,
  - *min_dist* $\in [0.15, 0.40]$.
- HDBSCAN:
  - *min_cluster_size* $mcs \in \{30, 40, \dots, 100\}$,
  - *min_samples* is tied to mcs (*min_samples* = *mcs*),
  - cluster_selection_method $\in \{$*eom*, *leaf*$\}$.

Tying *min_samples* to *min_cluster_size* reduces the dimensionality of the search and enforces a conservative density threshold: clusters must have both sufficient size and local density to be considered valid. This was motivated by earlier runs where independently small *min_samples* values produced many tiny, unstable topics and increased outlier sensitivity. Within each Optuna trial, we perform the following steps:

1. Fit the BERTopic model on the training set (GPU UMAP+HDBSCAN), compute training metrics, and apply *reduce_outliers*.
2. Transform the validation set and compute *pre-update* validation metrics (coherence, diversity, blended outlier rate, and mega-cluster share). These pre-update metrics are used to compute the objective J_{valid} returned to Optuna.
3. After computing J_{valid}, apply *update_topics* on the validation documents to obtain *post-update* topic descriptors. These post-update metrics are logged for diagnostic analysis only and do not influence hyperparameter selection.
4. Trials that collapse nearly all documents into outliers are pruned immediately to avoid unnecessary computation.

Trials that collapse nearly all documents into the outlier class are pruned immediately to avoid wasting computation. Although Optuna provides a median-based early-stopping rule for underperforming trials, we disable it here to observe the full behavior of the search space and avoid prematurely discarding configurations that may yield useful insights.

2.5. Baseline Model Selection and Behavior

The best baseline model selected by J_{valid} exhibits moderately high coherence with distinct, interpretable topics across brands. It produces a healthy number of topics with a low mega-cluster share, indicating that no single cluster dominates the corpus. Finally, the model maintains a non-trivial but manageable outlier rate after applying *reduce_outliers*, supporting stable downstream analysis.

Because BERTopic allows us to inspect *get_topic_info()* and representative tweets, we manually validated that the major topics capture intuitive brand narratives: e.g., game chatter and coaching discussions for Falcons, transit delays and safety concerns for MARTA, and food quality and service experience for Waffle House.

3 Real-Time Extensions of BERTopic

Performing model selection on a static dataset yields a strong baseline model, but Sprout Social's production environment requires models that can adapt to continuous streams of new data. Starting from our tuned baseline BERTopic model, we consider two incremental strategies for handling new tweets over time:

1. An **online** BERTopic variant that incrementally updates PCA, clustering, and vectorization via *partial_fit*.
2. A **merge-model** BERTopic approach that repeatedly fits local BERTopic models to new windows and merges them into a fixed baseline model.

We present the online BERTopic first, as it represents the most natural streaming-style extension of BERTopic given current library constraints. However, we ultimately adopt the merge-model strategy as our primary incremental method. As we show below, the online BERTopic faces structural limitations—including the need to pre-specify the number of clusters and the loss of non-linear manifold structure—that make it less suitable for crisis detection across many brands. Nevertheless, we include the online experiments for completeness and as a potential starting point for future engineering work at Sprout Social.

3.1. Online BERTopic

To evaluate a streaming-style extension of BERTopic, we implemented an incremental variant using IncrementalPCA, MiniBatchKMeans, and OnlineCountVectorizer, applying *partial_fit* across training and validation batches [8, 11]. We then compared topic structure on the validation set before and after incremental updates to assess stability.

3.1.1 Instability under Partial Fitting

Even with a fixed data split, the incremental model exhibited substantial structural drift. Several topics shrank or disappeared, while others absorbed hundreds of additional documents; for example, one topic grew from 122 to 767 documents after *partial_fit*. The topic-size histogram in Figure 3 shows that incremental updates cause major reallocations of documents across topics, reflecting significant drift in the underlying cluster structure.

3.1.2 Sources of Drift

This instability reflects the behavior of the underlying components. MiniBatchKMeans continuously updates cluster centroids, and OnlineCountVectorizer adjusts term statistics, causing both the geometric and lexical definitions of topics to shift after each batch [10]. Because BERTopic's *partial_fit* does not enforce semantic continuity, topic IDs are effectively redefined over time and topic identity is not preserved across updates.

3.1.3 Implications for Crisis Detection

Although the online variant demonstrates that maintaining a single evolving model is technically feasible, the lack of stable topic identities makes it unsuitable for longitudinal crisis monitoring, where persistent topic labels are essential for interpreting drift signals over time. Consequently, we treat the online model as an exploratory baseline and rely on the merge-model approach as our primary incremental strategy.

3.1.4 Design Trade-offs

Online BERTopic offers conceptual simplicity but introduces several structural limitations:

- Loss of non-linear manifold structure (UMAP) and density-based clustering (HDBSCAN).
- No explicit outlier detection; all points contribute to centroid updates.
- MiniBatchKMeans requires pre-specifying the number of clusters k, which does not scale well to thousands of brands with varying topical complexity.
- Centroid reassignment may overwrite or obscure low-volume yet important emerging topics.

These limitations conflict with the requirements of crisis detection, where rare but emerging negative narratives must be preserved rather than absorbed into majority clusters. While the online approach may be useful for lightweight streaming scenarios, it is not well aligned with the needs of large-scale, multi-brand crisis monitoring at Sprout Social.

3.2. Merge-Model BERTopic

Motivated by the limitations of the online BERTopic variant—particularly the need to pre-specify the number of clusters, the lack of explicit outlier handling, and the loss of non-linear manifold structure—we adopt BERTopic's *merge_models* pipeline as our primary incremental strategy.

3.2.1 Model Design

The merge-model approach operates as follows:

- A tuned static model G (trained on training set) serves as the *baseline*.
- Separate BERTopic models D_{valid} and $D_{\text{test\_fit}}$ are fit on new time windows using *the same* embedding, UMAP, and HDBSCAN parameters.
- Each D is merged into G using cosine similarity in c–TF–IDF space at a fixed threshold τ.

Because HDBSCAN assigns outliers and supports non-spherical clusters, delta models maintain high-quality topical structure even in small windows. The merge operation computes similarities between topics from the base and delta models in the c-TF–IDF space. When cosine similarity exceeds τ, topics are merged; otherwise, new topics are introduced into the baseline topic space. Intuitively, smaller τ values aggressively merge semantically similar topics, while large τ values preserve fine-grained distinctions and create more "new" topics. We focus on $\tau = 0.95$, which we find best preserves baseline topics while still allowing genuinely new topics to emerge.

Limitation — Document–topic remapping: A practical limitation of the current *merge_models* implementation is that it operates at the level of topic templates, not at the level of document assignments. After merging a baseline model G with a delta model D, BERTopic returns a new model with a unified set of topics, but it does not automatically provide a consistent mapping of *all* documents from all windows into this new topic index space.

For downstream brand and sentiment analysis, however, we require a global assignment:

- each tweet in training, validation, test-fit, and test-monitor sets should have a topic ID defined in the merged model, and
- those topic IDs must be consistent across time windows (so topic k means the same semantic concept before and after merging).

To address this, we adopt a re-inference strategy to enforce global consistency across all time windows. Specifically, we:

1. Aggregate the documents and pre-computed embeddings from all historical splits (training, validation, and test-fit) into a single dataset.
2. Apply the final merged model's *transform* pipeline to this full dataset, generating a unified set of topic assignments based on the merged topic centroids.
3. Refine these assignments using outlier reduction (HDBSCAN soft-clustering) to ensure that every non-noise tweet is mapped to the most relevant global topic in the unified space.

This additional layer is not provided out of the box by BERTopic and is a key engineering step in making merge-based modeling usable for change detection. It also highlights a conceptual limitation: after merging, documents from different time windows are effectively "projected" into the merged topic space indirectly through topic-level matches, not via a single global clustering pass over embeddings.

3.2.2 Delta HDBSCAN Tuning

Because validation and test-fit sets are smaller slices, we re-tune HDBSCAN for these "delta" models while holding the embedding and text parameters fixed. We perform a targeted Optuna study over:

- *min_cluster_size* (*mcs*),
- *min_samples* (again tied to *mcs*),
- and *cluster_selection_method*.

The objective is aligned with J but places slightly different emphasis on avoiding degenerate delta models with too few topics. We then use the best HDBSCAN configuration to fit D_{valid} and $D_{\text{test\_fit}}$.

3.2.3 Topic Stability and Change Metrics

To evaluate merge behavior, we construct *topic-change* tables comparing topic statistics *before* and *after* merging. For each topic we track:

- n_{before} and n_{after} (document counts),
- the change $n_{\Delta} = n_{\text{after}} - n_{\text{before}}$,
- deltas in sentiment distribution (per-topic changes in negative, neutral, positive share),
- deltas in brand distribution (per-topic changes in Waffle House/MARTA/Falcons proportions).

We then summarize:

- the proportion of *persisting* topics with small relative growth $|n_{\Delta}|/n_{\text{before}} < 0.10$;
- the proportion of persisting topics whose dominant sentiment stays constant;
- average absolute sentiment and brand shifts across persisting topics;
- the share of *new* topics, i.e., topics with $n_{\text{before}} = 0$ (denoted $\mathrm{prop\_delta\_new}$ in the code).

To focus on crisis-related structure, we also filter to persisting topics whose *after-merge* dominant sentiment is negative and compute the same metrics on this subset.

Empirically, we observe strong topic stability in the sense that:

- most baseline topics persist with modest size changes,
- dominant sentiment and brand composition are stable for the majority of persisting topics,
- a small but meaningful set of new negative topics emerges, which is exactly what a crisis detector needs to surface.

4 Change Detection on Merged Topics

Once we obtain a merged model M_{final}, we assign topics to all non-outlier tweets in the combined historical set and in test-monitor set. We then compute distribution-based drift metrics over daily batches.

4.1. Reference Distributions

We construct a *reference window* using all non-outlier tweets from the training, validation, and test-fit sets. From this window, we estimate the topic distribution $p_{\text{topic}}(k)$ over topic IDs, the sentiment distribution $p_{\text{sent}}(s)$ over $\{\text{negative}, \text{neutral}, \text{positive}\}$, the negative-topic distribution $p_{\text{neg-topic}}(k)$ among tweets with negative sentiment, and the overall negative prevalence μ_{neg} with its standard deviation σ_{neg} across days in the reference window.

These serve as baselines for the normal behavior of the brand ecosystem under the tuned model.

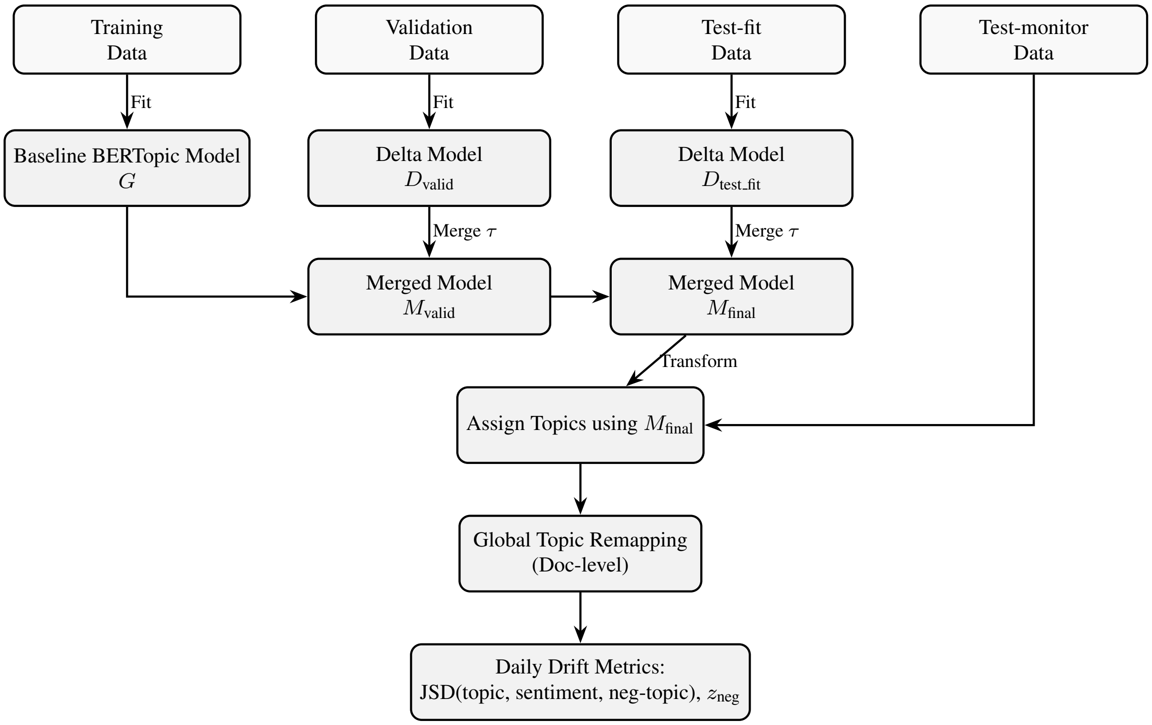

Figure 1. Overview of the merge-model BERTopic pipeline. A baseline model G is tuned and fit on training set; delta models fitted on validation and test-fit sets are merged via c–TF–IDF cosine similarity at threshold τ to produce M_{valid} and M_{final}. The final model assigns topics to test-monitor set, after which a global remapping layer produces a unified topic space for downstream drift detection.

4.2. Daily Batching and Drift Metrics

We allocate 70% of test set to test-fit and the remaining 30% (approximately 20,000 tweets) to test-monitor, so that the monitoring window contains enough volume for reliable estimation of daily topic and sentiment distributions and overall negative prevalence.

For test-monitor, we group tweets by calendar day and, for each day d, compute:

- daily topic distribution $q^{(d)}_{\text{topic}}$,
- daily sentiment distribution $q^{(d)}_{\text{sent}}$,
- daily negative-topic distribution $q^{(d)}_{\text{neg-topic}}$,
- daily negative prevalence $\hat{p}^{(d)}_{\text{neg}}$.

We quantify distribution shifts using Jensen–Shannon divergence (JSD) [13]:

$$
\begin{aligned}
\text{JSD}^{(d)}_{\text{topic}} &= \text{JSD}(q^{(d)}_{\text{topic}} \,\|\, p_{\text{topic}}),\\
\text{JSD}^{(d)}_{\text{sent}} &= \text{JSD}(q^{(d)}_{\text{sent}} \,\|\, p_{\text{sent}}),\\
\text{JSD}^{(d)}_{\text{neg-topic}} &= \text{JSD}(q^{(d)}_{\text{neg-topic}} \,\|\, p_{\text{neg-topic}}),
\end{aligned}
$$

where all distributions are reindexed to a joint support with missing probabilities set to zero.

To track overall negativity we compute a z-score:

$$
z^{(d)}_{\text{neg}} = \frac{\hat{p}^{(d)}_{\text{neg}} - \mu_{\text{neg}}}{\sigma_{\text{neg}}}.
$$

This yields four complementary signals:

- topic drift (do the relative frequencies of topics change?),
- sentiment drift (does the mix of negative vs. neutral vs. positive shift?),
- negative-topic drift (are negative tweets concentrated in different topics?),
- level shifts in negativity (does the overall fraction of negative tweets spike?).

4.3. Alert Heuristics and Visualization

We adopt a simple heuristic for identifying potential crisis days:

- For the topic JSD metric, we define a dynamic threshold: for each day d, we take the previous w days of $\text{JSD}_{\text{topic}}$ and set the threshold to $\max(\textit{alert\_floor}, \text{95th percentile of that window})$. In our experiments, we use $w = 5$ and $\textit{alert\_floor} = 0.20$.

- We then apply a hybrid rule: a day is treated as an alert candidate only if $\mathrm{JSD}_{\text{topic}}^{(d)}$ exceeds this dynamic threshold *and* the negative-prevalence z-score satisfies $z_{\text{neg}}^{(d)} > z_{\text{threshold}}$. We set $z_{\text{threshold}} = 2.5$.
- When the hybrid condition is met, the system prints a summary of the most negatively shifting topics (filtered by minimum negative share), supporting qualitative inspection of the underlying conversational changes.

Although the test-monitor set spans only the tail portion of the test period (approximately 20,000 tweets), it remains much smaller than the historical reference window and serves as a realistic approximation of a live monitoring segment. This setting allows us to test detector behavior under plausible operational conditions without requiring a full streaming deployment.

We also considered applying classical change-point detection methods to scalar series such as daily negative prevalence. However, given the short monitoring horizon and the inherently multi-dimensional nature of topic and sentiment dynamics, distribution-based JSD tracking provides a more robust and flexible framework across brands.

5 Results and Discussion

5.1. Baseline Model Performance

Tables 1 and 5 summarize quantitative metrics across the training, validation, and test splits, as well as qualitative examples of highly negative, crisis-relevant topics. Together, they demonstrate that the baseline model exhibits strong coherence, balanced diversity, controlled outlier rates, and interpretable negative themes that are brand-specific.

A key limitation of the baseline model is that it cannot introduce new topics when transforming unseen data. Because BERTopic does not re-run UMAP or HDBSCAN during *transform*, documents in the validation and test sets are assigned only to existing clusters or marked as outliers. Consequently, the topic inventories in the validation and test splits largely reuse the structure of the training split, with little opportunity for new emerging narratives to surface. This motivates the incremental extensions studied below, where the goal is to accommodate evolving conversational patterns while preserving stability for downstream monitoring.

5.2. Merge-Model Performance

Table 2 reports the number of non-outlier topics identified by the baseline model G, the validation delta model D_{valid}, and the test-fit delta model $D_{\text{test\_fit}}$. The validation slice yields a topic inventory slightly larger than the baseline, indicating that the tuned HDBSCAN parameters generalize reasonably well to a nearby time window. In contrast, the test-fit slice collapses to only 26 topics, reflecting the well-documented sensitivity of density-based clustering to local density variation and the poorer generalization of the tuned parameters on this particular window.

To understand how the merge procedure reconciles these distinct topic spaces, Table 6 summarizes merge outcomes across similarity thresholds $\tau \in \{0.85, 0.90, 0.95\}$. Lower thresholds aggressively merge incoming topics into existing ones, while higher thresholds preserve finer structure and introduce more new topics. At our final setting of $\tau = 0.95$, the merge recovers a substantial number of coherent new themes from both delta models—69 from the validation slice and 15 from the test-fit slice—despite the severe clustering collapse in the latter. This demonstrates that merge-model BERTopic can reconstruct topical granularity that standalone HDBSCAN would otherwise suppress.

Tables 7 and 8 illustrate representative new topics introduced during each merge. These include police-related incidents, Mother's Day content, Waffle House behavioral patterns, season preparation discussions, EV charging announcements, and civic or legal commentary. Such topics represent genuine conversational signals that do not appear in the static baseline and highlight the merge-model pipeline's ability to surface emerging narratives even when operating on challenging slices.

Stability Metrics for Persisting Negative Topics

We evaluate the stability of negative-dominant topics across merges along three axes: sentiment shift, brand composition shift, and document-count change. Aggregated results are shown in Tables 3 and 4 for the validation merge M_{valid} and the final merged model M_{final}.

For the validation merge, sentiment and brand shifts remain small for the majority of topics, as reflected in the tight clusters around zero in Figure 4. Document-count deltas in Figure 5 show that although most topics change modestly in size, several experience notable positive growth, highlighting emerging areas of user attention.

Top-changing topics: Tables 9 and 11 report the top persisting and top new negative topics ranked by size change in the validation merge.

Final-merge stability and delta-model collapse: After merging the test-fit window, the number of newly introduced topics decreases sharply, as shown in Table 12. However, this should not be interpreted as evidence that the topic space has fully stabilized. The delta model for the test-fit slice collapses to only 26 non-outlier topics and exhibits an extremely high mega-cluster share, as shown in Table 2. This collapse limits the granularity of the incoming topic structure, leaving the merge procedure with fewer opportunities to identify genuinely new themes. Thus, the small

Table 1. Baseline BERTopic model metrics on training, validation, and test splits. For the validation and test splits, entries in the form "a / b" indicate pre- and post-*reduce_outliers* values.

Metric	Train	Valid (Pre/Post)	Test (Pre/Post)
Coherence	0.5903	0.4355 / 0.5475	0.5056 / 0.5628
Diversity	0.3273	0.3273 / 0.6069	0.3273 / 0.5956
Redundancy (Jaccard)	0.0105	0.0105 / 0.0087	0.0105 / 0.0088
Outlier rate (before/after)	0.8106 / 0.1759	0.8271 / 0.2057	0.8434 / 0.1741
Mega-cluster share	0.0399	0.0600	0.0744
Number of topics	212	202	203
J score	0.4705	0.4051	0.4287

Table 2. Non-outlier topic counts in the baseline (G), validation delta (D_{valid}), and test-fit delta ($D_{\text{test\_fit}}$) models. The strong collapse in the test-fit slice demonstrates the sensitivity of GPU UMAP+HDBSCAN to local density structure.

Model	Non-outlier Topics
Baseline G	212
D_{valid}	237
$D_{\text{test\_fit}}$	26

Table 3. Stability metrics for persisting negative-dominant topics in M_{valid}.

Metric	Value
avg abs sentiment shift	0.01998
max abs sentiment shift	0.11739
avg abs brand shift	0.00841
topics with stable dominant brand	72/72
topics with small doc change	10/72
num new topics	16

Table 4. Stability metrics for persisting negative-dominant topics in M_{final}.

Metric	Value
avg abs sentiment shift	0.00469
max abs sentiment shift	0.03903
avg abs brand shift	0.00217
topics with stable dominant brand	86/86
topics with small doc change	64/86
num new topics	6

number of newly introduced topics in M_{final} reflects limited structure in $D_{\text{test\_fit}}$ rather than convergence in the underlying topic landscape.

Figures 6 and 7 illustrate sentiment–brand shifts and document-count deltas for M_{final}. While the shifts cluster more tightly around zero, this apparent stability should be interpreted alongside the degenerate structure of the test-fit delta model.

Tables 10 and 12 summarize the highest-shifting and newly introduced negative topics after the final merge.

Taken together, these results show that the merge-model pipeline produces a topic space that is both *adaptive*—able to integrate meaningful emerging themes from additional windows—and *stable* for persisting topics when the incoming delta window contains sufficient structure. Even when delta models collapse under density-based clustering, merge-model BERTopic is able to recover a granular and semantically rich topic landscape, providing a robust foundation for downstream drift detection.

5.3. Change-Detection Behavior

Using the final merged model M_{final}, we apply the drift detector introduced in Section 4 to the test-monitor window. Figure 2 plots the daily Jensen–Shannon divergences between the monitoring distributions and the historical reference: topic distributions, sentiment distributions, and negative-topic distributions. The figure also shows the dynamic threshold for topic JSD derived from a five-day rolling window.

Topic JSD fluctuates between approximately 0.10 and 0.21, generally staying below the dynamic threshold. Sentiment JSD remains near zero throughout, indicating that aggregate brand-level sentiment remains stable. Negative-topic JSD (red) exhibits slightly larger day-to-day variation, reflecting the natural shifting of which specific negative topics are active; however, these changes are gradual rather than abrupt. Taken together, the three divergence curves are consistent with a relatively calm period: the conversational mix drifts incrementally but does not deviate sharply from the historical baseline.

To convert these continuous signals into discrete alert candidates, we apply the hybrid rule described earlier: an alert is raised on day d only when (i) $\text{JSD}^{(d)}_{\text{topic}}$ exceeds its dynamic threshold and (ii) the negative-prevalence z-score $z^{(d)}_{\text{neg}}$ exceeds 2.5. Table 13 reports the resulting hybrid metrics for each monitoring day.

Across the monitoring window, no day satisfies both criteria simultaneously. Topic divergence never exceeds the dynamic threshold by a substantial margin, and negative-prevalence z-scores remain within a narrow band around zero (approximately ± 0.3). As shown in Table 13, neither metric exhibits the type of joint spike associated with crisis-like shifts.

In this setting, the system functions as a low–false-positive detector on a quiet period of data. In future work with labeled crisis events or synthetic perturbations, the same hybrid framework could be evaluated for its ability to detect known incidents and stress-tested under more volatile conditions.

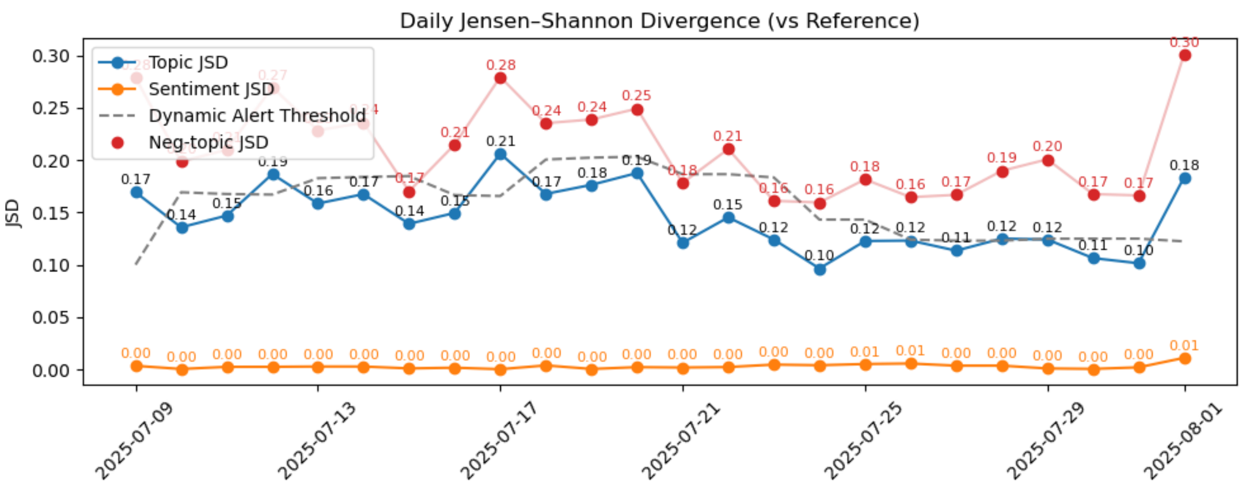

Figure 2. Daily Jensen–Shannon divergence between test-monitor and the reference window. The dashed line denotes the dynamic topic JSD alert threshold computed from a rolling window of recent days. An alert is raised only when both the topic JSD exceeds this threshold and the negative-prevalence z-score is statistically high.

6 Limitations and Future Work

6.1. Scale and Parameter Sensitivity

Our experiments focus on three brands and one year of data. Scaling this pipeline to Sprout Social's full ecosystem of 30,000+ brands raises several open questions:

- **Topic-capacity scaling:** How should the number of topics (whether controlled through HDBSCAN density thresholds or k for k-means) scale with total tweet volume, brand heterogeneity, and seasonal variability?

- **Adaptive alert thresholds:** How should JSD thresholds and z_{neg} thresholds be adapted for brands with extremely high volume (stable distributions) versus sparse brands where day-to-day fluctuations dominate?

- **Robustness of tuned hyperparameters:** To what extent do our tuned parameters (e.g., UMAP dimensionality, HDBSCAN *min_cluster_size*, *min_samples*, and similarity threshold τ) transfer to new brands or domains exhibiting different linguistic patterns or crisis dynamics?

In the online BERTopic variant, specifying k is particularly challenging: naively scaling the number of clusters by volume is likely to overestimate topic capacity when brands share overlapping thematic structure (e.g., IHOP vs. Waffle House). This suggests future work on hierarchical, brand-aware clustering or dynamic topic splitting/merging architectures.

6.2. Reproducibility of GPU-Accelerated UMAP and HDBSCAN

A second limitation lies in the reproducibility of the GPU-accelerated manifold learning and clustering components. Conceptually:

- **UMAP** is stochastic: it initializes embeddings randomly and optimizes a non-convex objective through stochastic updates. Even with fixed seeds, slight numerical differences across GPUs or library versions can yield different embeddings.

- **HDBSCAN** is deterministic given an exact distance matrix and fixed hyperparameters, but its output depends on the k-nearest neighbor graph produced upstream, which itself depends on UMAP and GPU stochasticity.

In cuML, both UMAP and HDBSCAN leverage GPU parallelism and, in some cases, approximate neighbor search, introducing additional sources of non-determinism. In practice, we observed that re-running the same configuration (same data, hyperparameters, and seeds) can yield small variations in embeddings and cluster assignments. While these differences are usually minor, they can affect topics near decision boundaries, occasionally causing cluster splits, merges, or shifts in outlier assignments.

Empirically, we found:

- BERTopic is highly sensitive to manifold and clustering hyperparameters [3],

- and small stochastic variations can push the model from a "good" region (coherent, balanced topics) to a "fragile" region (mega-clusters or many tiny topics).

To manage these issues, we:

- used Optuna to explore hyperparameter regions exhibiting local robustness across folds and slices,
- persisted best-performing models to disk rather than relying on run-to-run determinism,
- and emphasized aggregate metrics (coherence, diversity, outlier rate, mega-cluster share) and qualitative topic inspection over strict run-to-run identity of topic IDs.

These reproducibility considerations do not preclude deployment of GPU-accelerated BERTopic, but they must be acknowledged—particularly when comparing experiments or performing automated regression testing.

6.3. Limitations of the Merge-Model Approach

While the merge-model strategy provides a practical way to extend a tuned baseline BERTopic model over time, it introduces several conceptual and operational limitations:

No global re-clustering of embeddings: *merge_models* joins topics solely through cosine similarity in c–TF–IDF space; it does not re-run UMAP or HDBSCAN on the union of embeddings. Thus:

- topic boundaries remain anchored to the geometry of the baseline model G [6], and
- new themes can create new topics or merge into existing ones, but cannot reshape global structure.

This increases stability but also means that any imperfect baseline decisions (e.g., an early merged topic that was too broad) persist indefinitely.

Sensitivity to the similarity threshold τ: The threshold τ controls when topics from a delta model are merged into the baseline. If τ is too low, semantically distinct topics merge, reducing interpretability. If τ is too high, the model proliferates new topics, making downstream analysis noisy. We found $\tau = 0.95$ worked well on our dataset, but this is heuristic and dataset-dependent.

Accumulated complexity with repeated merges: As additional windows are merged (e.g., baseline $\rightarrow$ validation $\rightarrow$ test-fit $\rightarrow$ future slices), the topic space can accumulate:

- many small, low-support topics introduced in later windows,
- subtle representational drift across generations of merges.

Our global remapping layer ensures consistent document-level topic IDs, but semantically similar topics may remain fragmented if they never exceed the τ threshold at merge time.

Baseline bias and rare-topic fragility: Because G acts as the anchor, high-volume patterns in early data can dominate the topic space. Later rare but important patterns (e.g., niche safety crises) form their own topics only if they generate sufficient mass in a delta model *and* pass the merge threshold. Otherwise, they may be absorbed into high-volume topics, weakening change-detection sensitivity.

Interaction with stochastic GPU components: The merge-model approach inherits stochastic GPU variation from UMAP and HDBSCAN. Small shifts in embeddings can influence which topics surpass the similarity threshold τ, affecting merge outcomes. This further motivates hyperparameter tuning for robustness and treating topic IDs as approximate identifiers rather than fixed semantic units.

Despite these limitations, the merge-model pipeline offers a practical balance of stability, interpretability, and engineering feasibility for our three-brand prototype. These limitations highlight critical areas for future refinement at Sprout Social's full production scale, where mis-merged or fragmented topics could affect automated monitoring.

6.4. Crisis Labeling and Evaluation

Our evaluation is currently unsupervised, as we do not have ground-truth crisis labels. As a result, we cannot compute precision, recall, or false-alarm rates for alerts. Future work should:

- Construct annotated case studies in which known crises are aligned with social media timelines and labeled at appropriate temporal granularity.
- Test whether our drift metrics (topic JSD, sentiment JSD, negative-topic JSD, and negative-prevalence z-scores) spike at the correct times and whether the surfaced topics align with human judgment.
- Compare our hybrid JSD + z-score detector against classical change-point detection methods applied to derived scalar series (e.g., daily negative prevalence in crisis-relevant topics).

7 Conclusion

We developed a BERTopic-based pipeline for social media crisis detection on a realistic multi-brand dataset. A tuned, GPU-accelerated baseline model provides coherent and diverse topics with explicit outlier handling. Using *merge_models*, we extend this baseline with new time windows while preserving topic identities and introducing interpretable new negative topics.

Our results show strong generalization on the validation slice, while the test-fit delta model collapses to a small number of clusters—highlighting the sensitivity of density-based clustering to distribution shift. The merge step still recovers new and meaningful topics, but this behavior indicates that incremental updates alone may not always maintain modeling capacity. For this reason, periodic full retraining on accumulated data is advisable, even if performed infrequently.

On top of the merged topic space, our drift detector—based on Jensen–Shannon divergence and negative-prevalence z-scores—provides stable, interpretable signals over daily batches. The online BERTopic variant further clarifies the trade-offs between streaming-style updates and loss of manifold structure, reinforcing the merge-model approach as the more reliable path forward.

Overall, transformer-based topic modeling combined with distributional drift metrics offers a practical foundation for crisis monitoring at Sprout Social's scale, while leaving open questions around robustness, large-scale deployment, and retraining strategy.

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A Appendix A: Engineering Notes on Merge-Model Topic Remapping

Why remapping is required: BERTopic's *merge_models* function operates at the *topic-template* level: it merges topics from a baseline model G and a delta model D based on cosine similarity between their c–TF–IDF representations (topics are merged when similarity exceeds a threshold τ). However, the API does not automatically remap the underlying *documents* of G or D into the unified topic index space of the merged model M.

For downstream crisis detection, we require:

- a *single, global* set of topic IDs valid across training, validation, test-fit, and test-monitor sets;
- document-level topic assignments that respect the merges discovered by *merge_models*.

Thus a custom remapping procedure is required.

Topic-level mapping: From *merge_models*, we extract a mapping:

$$(\textit{source\_model},\ \textit{local\_topic\_id}) \rightarrow \textit{merged\_topic\_id}.$$

This is derived directly from BERTopic's cosine-similarity evaluation of c–TF–IDF topic vectors:

$$\cos(\theta) = \frac{\mathbf{v}_G \cdot \mathbf{v}_D}{\|\mathbf{v}_G\|\ \|\mathbf{v}_D\|}.$$

When $\cos(\theta) \geq \tau$, the delta topic is merged into the nearest baseline topic; otherwise a new topic is created.

Global remapping table: Because merging can occur repeatedly (e.g., training + validation $\rightarrow M_{\text{valid}}$, then M_{valid} + test-fit $\rightarrow M_{\text{final}}$), we recursively compose these topic-level mappings to build:

$$\textit{global\_map} : (\textit{model\_name},\ \textit{local\_topic}) \rightarrow \textit{final\_topic\_id}.$$

This table uniquely identifies every topic instance across all intermediate models.

Document-level remapping: We apply *global_map* to the stored topic assignments of every document:

$$(\textit{doc\_id},\ \textit{local\_topic}) \mapsto (\textit{doc\_id},\ \textit{global\_topic}).$$

This yields a unified document–topic table with:

- a stable final topic ID,
- associated sentiment and brand labels,
- and timestamps for drift computation.

CSV outputs and reproducibility: Our pipeline exports:

- *topic_change_*.csv*: topic-level before/after merging statistics,
- *topic_assignments_merged.csv*: unified document-level topic assignments,
- *daily_drift_*.csv*: per-day JSD values, negative-prevalence z-scores, and alert flags.

These files are sufficient for a reader to:

1. reconstruct the merged topic space,
2. trace how topics evolve or appear over time, and
3. reproduce all drift metrics and alert conditions without re-running BERTopic.

Summary: Because BERTopic's merge operation produces only topic-template mappings, document-level consistency across time windows requires an explicit remapping layer. Our implementation—based on cosine-similarity merges in c–TF–IDF space—provides the necessary global topic space for change detection.

B Appendix B: Baseline and Merge-Model Topic Tables

This appendix collects topic-level tables referenced in Sections 5.2 and 5.3.

C Appendix C: Stability and Change-Detection Tables and Figures

This appendix collects stability metrics and hybrid alert tables and figures referenced in Sections 5.2 and 5.3.

Table 5. Examples of highly negative crisis-relevant topics from the baseline model (with explicit language removed). These topics have high negative sentiment and strong brand purity, capturing sentiment-intense narratives that are relevant for crisis monitoring.

Topic ID	**Neg**	**Neu**	**Pos**	**# Docs**	**Top Words**	**Doc. Share**	**Falcons**	**MARTA**	**Waffle**
65	0.966	0.034	0.000	58	franchise, franchises, poverty, league, years, ...	0.0011	0.8793	0.0000	0.1207
91	0.929	0.071	0.000	28	losing, btw, team, us, swept, lost, twice, ...	0.0005	0.9286	0.0000	0.0714
14	0.831	0.148	0.021	189	falcons, atlanta, fan, fans, playoffs, ...	0.0035	0.9947	0.0000	0.0053
132	0.829	0.143	0.029	35	playoffs, still, championship, losing, win, ...	0.0006	1.0000	0.0000	0.0000
128	0.793	0.155	0.052	58	fans, fan, base, fanbase, dumb, frenchy, ...	0.0011	0.9138	0.0000	0.0862
84	0.786	0.214	0.000	14	defense, horrible, kweer, smear, killa, mhm, ...	0.0003	0.9286	0.0000	0.0714
167	0.785	0.147	0.068	191	falcons, atlanta, fan, franchise, team, fans, ...	0.0035	0.9843	0.0000	0.0157
109	0.785	0.151	0.065	93	28, lead, super, qb, bowl, game, team, choke, ...	0.0017	1.0000	0.0000	0.0000

Table 6. Effect of cosine-similarity merge threshold τ on model size. Higher τ values introduce more new topics from delta windows.

τ	$\lvert G\rvert$	$\lvert D_{\text{valid}}\rvert$	$\lvert D_{\text{test\_fit}}\rvert$	$\lvert M_{\text{valid}}\rvert$	$\lvert M_{\text{final}}\rvert$	New from Valid	New from Test-fit
0.85	212	237	26	213	213	1	0
0.90	212	237	26	214	215	2	1
0.95	212	237	26	281	296	69	15

Table 7. Examples of new topics introduced during the first merge ($G \leftarrow D_{\text{valid}}$) at $\tau = 0.95$.

Topic ID	**Count**	**Name**	**Representation**	**Representative Doc**
271	389	275_son_his_dad_prank	[son, his, dad, prank, sanders, apology, ...]	cmon the actions of the son of one of the coaches...
272	155	272_mother_mom_day_mothers	[mother, mom, day, mothers, plaqueboymax, ...]	waffle house on mother's day l's up and cards...
273	27	273_hawks_valley_rbi_ip	[hawks, valley, rbi, ip, ks, tough, final, ...]	final from yesterday: hawks remain perfect in...
279	156	279_station_marta_police_shooting	[station, marta, police, shooting, officer, ...]	man killed in shooting near west end marta...
280	148	280_she_thought_thinks_was	[she, thought, thinks, was, her, at, house, ...]	she must have thought she was at a waffle house...

Table 8. Examples of new topics introduced during the second merge ($M_{\text{valid}} \leftarrow D_{\text{test\_fit}}$) at $\tau = 0.95$.

Topic ID	**Count**	**Name**	**Representation**	**Representative Doc**
286	611	286_help_support_dm_message	[help, support, dm, message, your, me, ...]	inbox me with your cash app or paypal...
287	204	287_butter_oil_real_margarine	[butter, oil, real, margarine, oils, tallow, ...]	real butter only please. i think y'all only have...
288	1628	288_season_wait_ready_year	[season, wait, ready, year, game, next, ...]	i can't wait for the season to begin!! let's go...
293	86	292_ev_charging_bp_pulse	[ev, charging, bp, pulse, fast, chargers, ...]	waffle house is getting fast ev charging...
295	115	295_court_filed_regina_shanita	[court, filed, regina, shanita, case, judicial...]	i didn't win my case and my pua benefits...

Table 9. Top 5 persisting negative-dominant topics by absolute size change after the validation merge.

topic_id	n_{before}	n_{after}	n_{Δ}	sent_neg	Representation
0	7063	23383	16320	0.480	[you, your, it, this, that, ...]
1	5232	9583	4351	0.417	[he, him, his, guy, ball, ...]
9	10046	12782	2736	0.528	[never, waffle, house, been, ...]
15	3584	5983	2399	0.497	[marta, bus, train, station, ...]
91	1222	3022	1800	0.771	[lose, lost, win, loss, game, ...]

Table 10. Top 5 persisting negative-dominant topics by absolute size change after the final merge.

topic_id	n_{before}	n_{after}	n_{Δ}	sent_neg	Representation
9	12782	15638	2856	0.532	[never, waffle, house, been, ...]
6	6603	8322	1719	0.465	[she, her, waitress, house, ...]
0	23383	25091	1708	0.490	[you, your, it, that, bro, ...]
62	6662	8154	1492	0.366	[me, my, waitress, her, ...]
78	3167	4099	932	0.880	[you, your, re, back, go, ...]

Table 11. Top 5 new negative-dominant topics (by n_{after}) introduced in the validation merge.

topic_id	n_{after}	sent_neg	Representation
212	630	0.508	[ulbrich, sanders, shedeur, prank, son, ...]
266	592	0.856	[black, women, videos, woman, ...]
213	493	0.767	[ulbrich, jeff, son, jax, ...]
221	475	0.674	[jasmine, crockett, mace, nancy, ...]
270	434	0.876	[son, coach, information, kid, ...]

Table 12. Top 5 new negative-dominant topics (by n_{after}) introduced in the final merge.

topic_id	n_{after}	sent_neg	Representation
281	653	0.868	[trump, maga, dirtybirds, felon, rw, ...]
294	241	0.481	[vick, mike, michael, dog, dogs, ...]
283	199	0.794	[dogs, dog, killer, animals, ...]
285	181	0.591	[gas, prices, inflation, low, gallon, ...]
295	115	0.739	[court, filed, regina, shanita, case, ...]

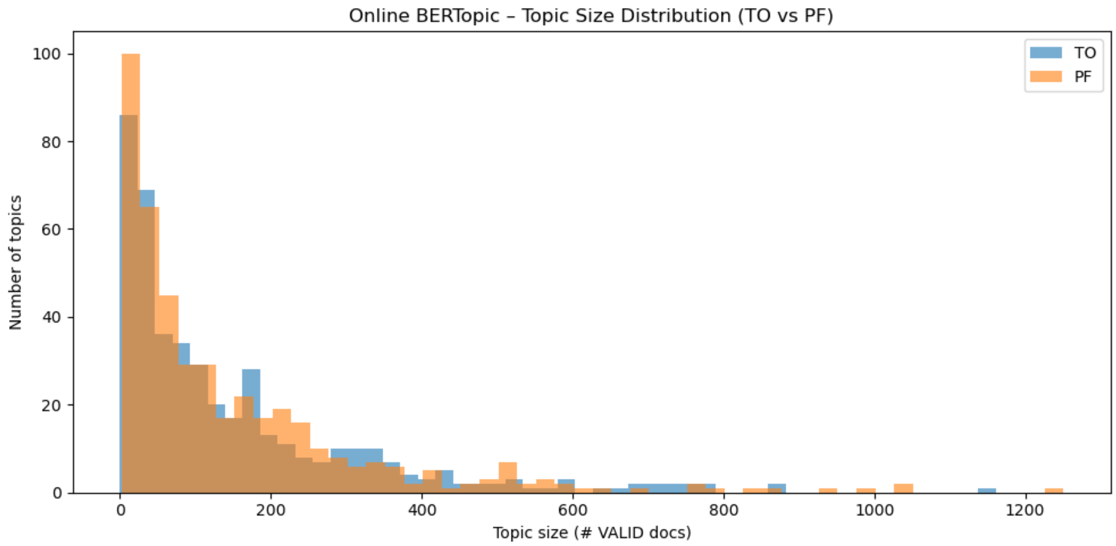

Figure 3. Topic size distribution before (TO) and after (PF) applying *partial_fit* on validation data. This histogram shows how *partial_fit* reshapes topic-document assignments across batches.

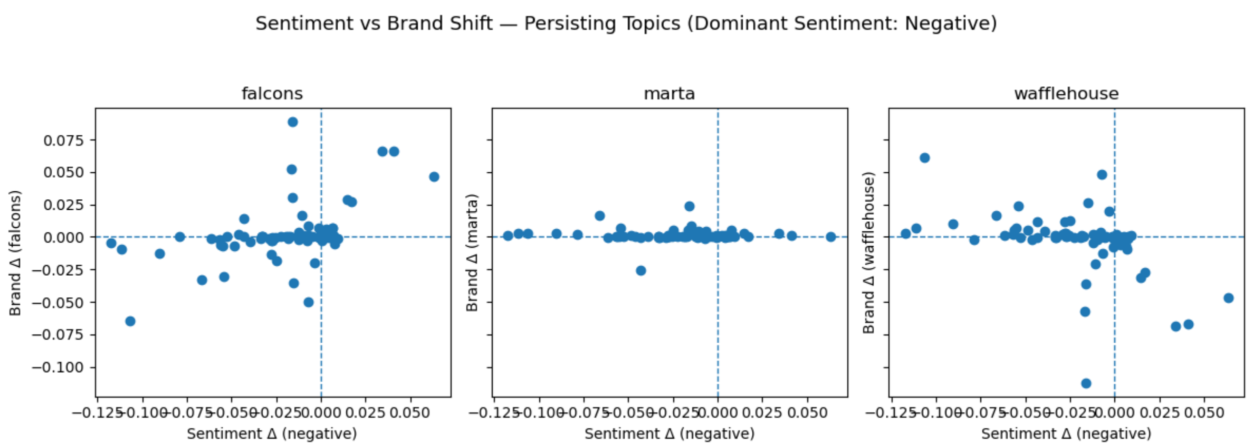

Figure 4. Sentiment–brand shift scatterplots for persisting negative-dominant topics in M_{valid}. Shifts are small and centered around zero.

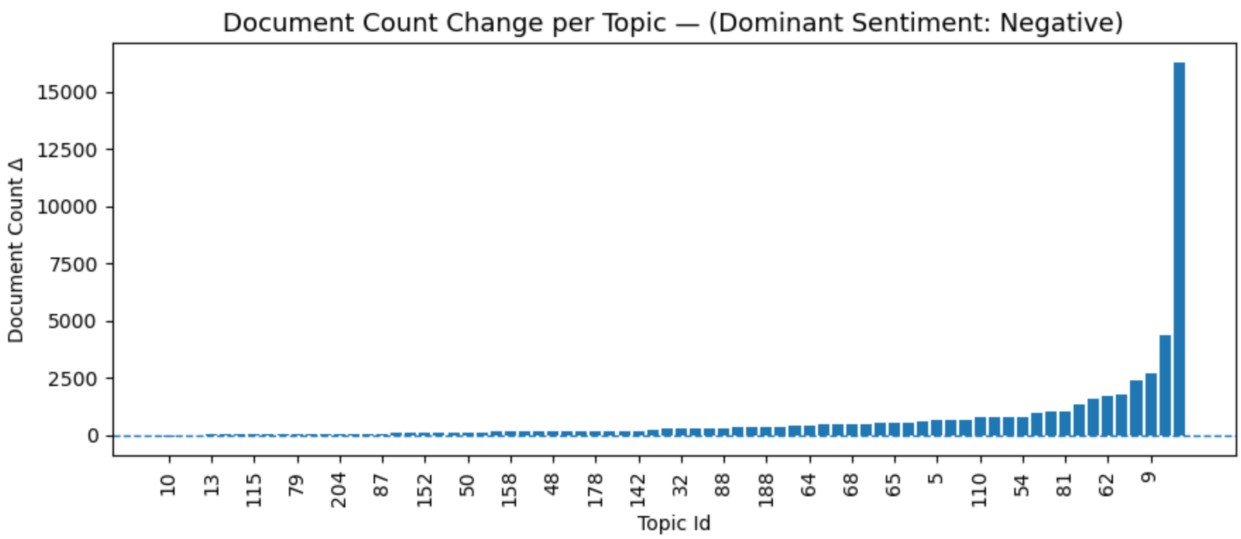

Figure 5. Document-count change per topic for persisting negative-dominant topics in M_{valid}. Most topics have modest changes, with a few showing substantial growth.

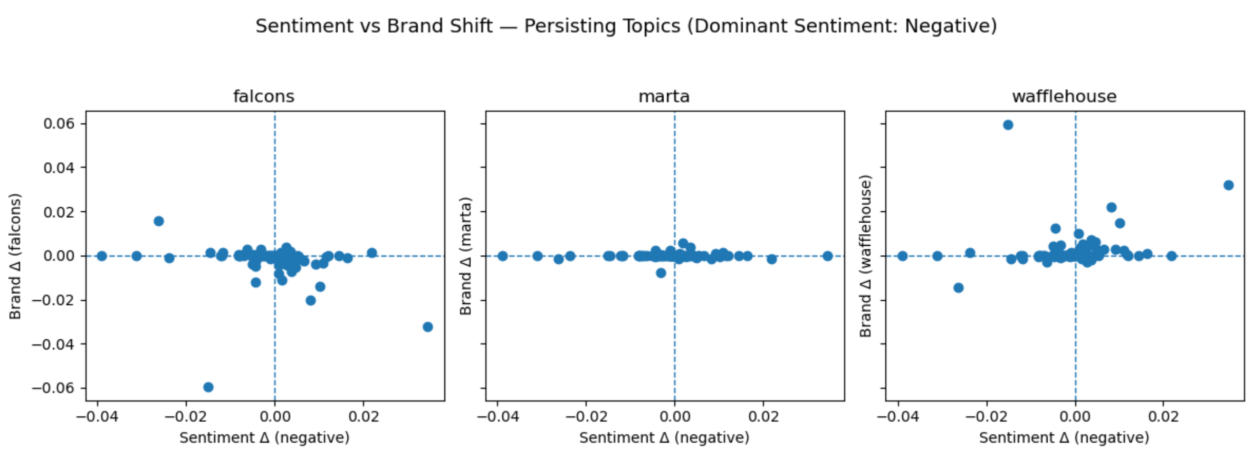

Figure 6. Sentiment–brand shift scatterplots for persisting negative-dominant topics in M_{final}. Shifts are minimal.

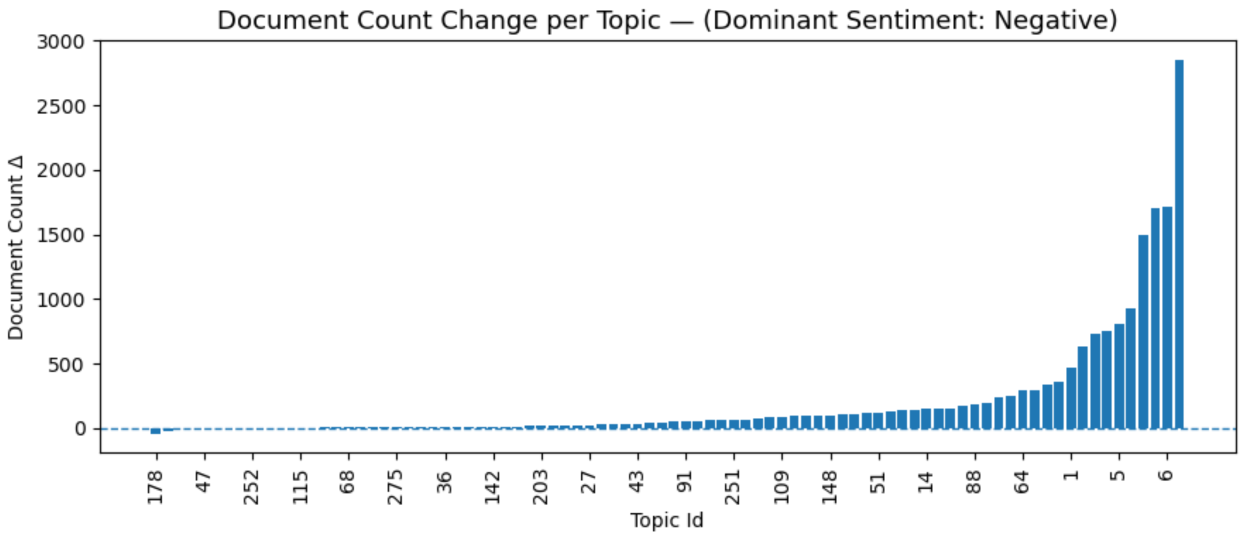

Figure 7. Document-count change per topic for persisting negative-dominant topics in M_{final}. Most topics show stable behavior with moderate growth in a few clusters.

Table 13. Hybrid alert evaluation over the test-monitor period. All days are ultimately classified as "OK" because topic JSD remains close to or below the dynamic threshold, and negative prevalence stays within roughly 0.3 standard deviations of the historical mean.

Date	Topic JSD	Threshold	Sentiment JSD	Neg. Prevalence	z_{neg}	Alert
2025-07-09	0.169	0.200	0.003	0.288	-0.16	OK
2025-07-10	0.136	0.200	0.000	0.341	-0.05	OK
2025-07-11	0.147	0.200	0.002	0.300	-0.13	OK
2025-07-12	0.187	0.200	0.002	0.298	-0.13	OK
2025-07-13	0.158	0.200	0.003	0.297	-0.14	OK
2025-07-14	0.167	0.200	0.003	0.294	-0.14	OK
2025-07-15	0.139	0.200	0.001	0.346	-0.03	OK
2025-07-16	0.149	0.200	0.002	0.315	-0.10	OK
2025-07-17	0.206	0.200	0.000	0.376	0.03	OK
2025-07-18	0.168	0.200	0.004	0.282	-0.17	OK
2025-07-19	0.176	0.200	0.000	0.337	-0.05	OK
2025-07-20	0.188	0.200	0.002	0.318	-0.09	OK
2025-07-21	0.121	0.202	0.002	0.309	-0.11	OK
2025-07-22	0.145	0.202	0.002	0.312	-0.11	OK
2025-07-23	0.124	0.200	0.005	0.275	-0.18	OK
2025-07-24	0.096	0.200	0.004	0.282	-0.17	OK
2025-07-25	0.123	0.200	0.005	0.276	-0.18	OK
2025-07-26	0.123	0.200	0.006	0.270	-0.19	OK
2025-07-27	0.113	0.200	0.004	0.320	-0.09	OK
2025-07-28	0.125	0.200	0.004	0.286	-0.16	OK
2025-07-29	0.124	0.200	0.001	0.353	-0.02	OK
2025-07-30	0.106	0.200	0.000	0.386	0.05	OK
2025-07-31	0.101	0.200	0.002	0.305	-0.12	OK
2025-08-01	0.184	0.200	0.011	0.230	-0.28	OK